

# Synthetic website replication tasks: diversity, quality, and agent behavior

This document shows the task set as an environment design artifact: generated multi-page websites, screenshot-only agent inputs, hidden oracle manifests, and browser-state grading. The focus here is not the reward internals; it is whether the generated tasks are varied, realistic, and useful for observing how coding agents reproduce website designs.

**14**

**packaged Harbor tasks**

All with hidden oracle sites and screenshot manifests.

**187**

**public reference screenshots**

Full pages plus meaningful states, exposed to the agent as numbered images.

**2**

**animation-marked tasks**

Marine biology and gemology include V1 animation captures.

**Core claim** The task set is diverse not only by topic, but by page grammar: dashboards, catalogs, booking flows, financial calculators, sports standings, scientific profiles, editorial music pages, maps, filters, tabs, accordions, and selected states.

# 1. Task atlas: generated, not crawled

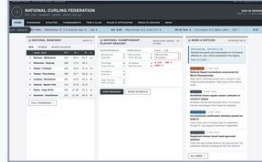
Each task is a synthetic oracle website generated from scratch, then frozen into screenshots and hidden replay metadata. This avoids crawling existing websites while still creating realistic browsing surfaces.

**Botanical Garden Nursery**



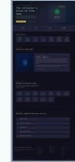
15 screenshots · static

**Curling Federation**



11 screenshots · static

**Gemology Reference Database**



17 screenshots · animation

**Indie Record Label**



11 screenshots · static

**Lutherie Guild**



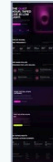
9 screenshots · static

**Marine Biology Reference Portal**



8 screenshots · animation

**Podcast Network**



17 screenshots · static

**Regional Credit Union**



12 screenshots · static

**Education K12**



18 screenshots · static

**Science Research Institute**



11 screenshots · static

**Retail Marketplace Vintage**



11 screenshots · static

**Storm Chasers Community**



10 screenshots · static

**Tabletop Gaming Cafe**



17 screenshots · static

**Zen Center**



11 screenshots · static

First reference screenshot for each of the 14 packaged tasks. Counts shown are public screenshots available to the agent.

## 2. Diversity overview

| Task                               | Domain                | Screenshots | Animation | Why it adds useful variety                                                        |
|------------------------------------|-----------------------|-------------|-----------|-----------------------------------------------------------------------------------|
| botanical-garden-nursery           | garden / nursery      | 15          | No        | Plant retail, visiting, membership, garden map, workshops, seasonal content.      |
| curling-federation                 | sports federation     | 11          | No        | Standings, results, brackets, athlete directory, rules, clubs.                    |
| gemology-reference-database        | reference database    | 17          | Yes       | Dense educational/reference pages: minerals, cuts, glossary, grading, provenance. |
| indie-record-label                 | music / label         | 11          | No        | Editorial/zine branding, releases, artists, catalog, tour, shop.                  |
| lutherie-guild                     | craft / guild         | 9           | No        | Artisanal workshop, materials, apprenticeship, exhibitions, journal pages.        |
| marine-biology-reference-portal    | science reference     | 8           | Yes       | Taxonomy, habitat, conservation, field guide, publication-heavy pages.            |
| podcast-network                    | media network         | 17          | No        | Episode archive, channel filters, audio/player surfaces, live events.             |
| regional-credit-union              | financial services    | 12          | No        | Calculators, sign-in flows, governance, branch finder, reading room.              |
| site-01-education-k12              | K-12 education        | 18          | No        | Institutional dashboard, lunch/calendar/board/academics/portal states.            |
| site-04-science-research-institute | research institute    | 11          | No        | People, publications, data, expeditions, careers, newsroom.                       |
| site-05-retail-marketplace-vintage | vintage marketplace   | 11          | No        | Browse, lot detail, cart, account, picker, activity log.                          |
| storm-chasers-community            | weather community     | 10          | No        | Ops dashboards, logs, spotters, gear, gallery, roster, events.                    |
| tabletop-gaming-cafe               | hospitality / gaming  | 17          | No        | Menu, booking steps, events, game library, reviews, membership.                   |
| zen-center                         | nonprofit / community | 11          | No        | Schedule, teachers, visits, audio, donations, application flows.                  |

### 3. What the agent generally does

#### Strengths

- Usually creates the expected page count and route scaffold.
- Often captures global art direction: palette, mood, nav, hero/footer.
- Builds recognizable components: cards, grids, menus, dashboards, filters, forms.
- Can reproduce strong static views when the page grammar is obvious.

#### Failure modes

- Card density, spacing, and row/column geometry drift.
- Stateful views are missed or only cosmetically implied.
- Rendered text/content structure changes even when the page looks plausible.
- Custom diagrams, maps, rosters, brackets, and inventory walls are approximated.

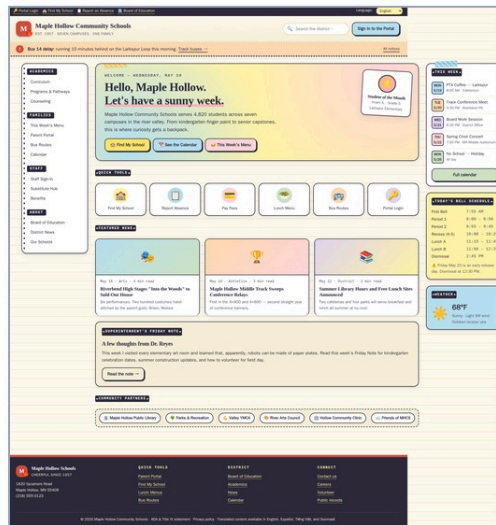
**Why this is useful for RL** The task set produces a spectrum rather than a binary outcome. Good candidates preserve structure and states; medium candidates reproduce the vibe but miss local details; weak candidates look plausible while failing layout/content fidelity.

# 4. Strong reproduction: institutional dashboard

## High-fidelity institutional dashboard

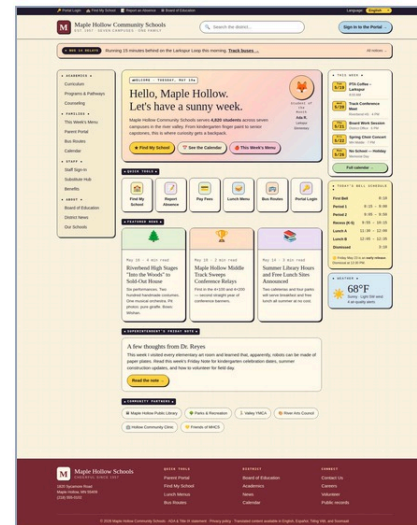
The agent preserves school dashboard structure, repeated cards, side rail, and dense content.

### Reference / oracle



trial reward 0.678 · coverage 0.957 · HTML 0.634 · CSSOM 0.646 · visual block 0.855  
Good reproductions keep page grammar, not just color and broad layout.

### Candidate



The K-12 task is a good high-end example: the agent keeps the dashboard grammar, side navigation, cards, footer, and school-specific density.

reward 0.678

coverage 0.957

HTML 0.634

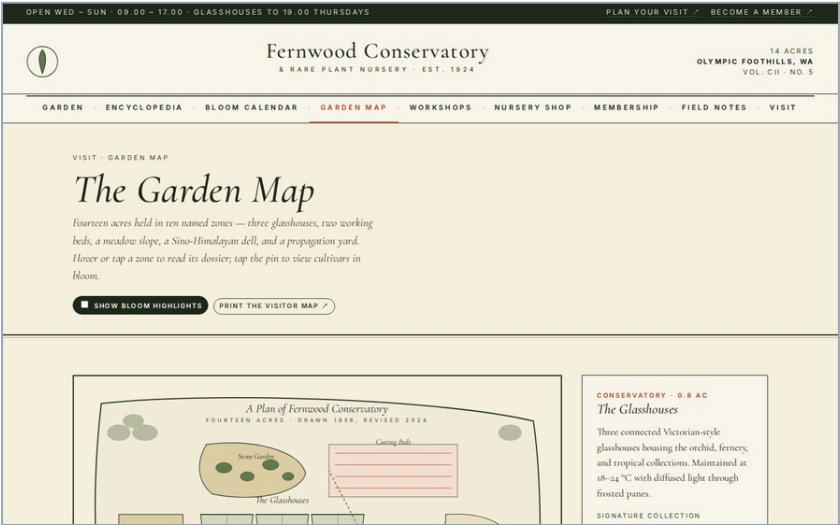
CSSOM 0.646

# 5. Strong state reproduction: botanical map

## Stateful botanical map reproduction

Map/highlight state shows the benchmark is not only static homepages.

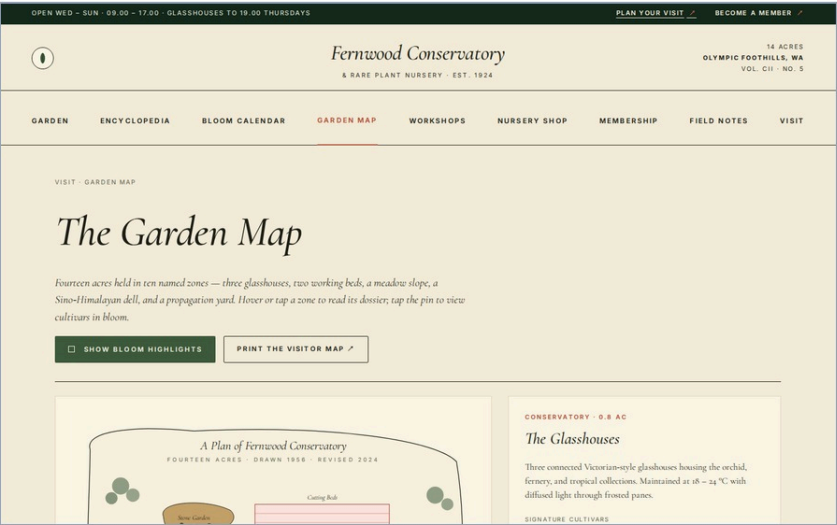
### Reference / oracle



capture score 0.810 · VB 0.951 · VB size 0.992 · block PM 0.606 · CSSOM 0.863

The candidate captures the same visible block system and highlighted state.

### Candidate



This task demonstrates that the dataset is not just homepage screenshots. The highlighted garden-map state requires preserving a specific browser-visible state.

capture score 0.810

visual block 0.951

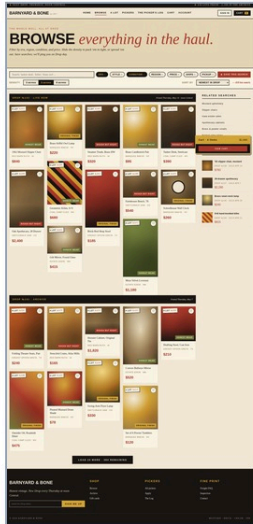
CSSOM 0.863

# 6. Commerce task: vintage marketplace

## Vintage marketplace with commerce density

Commerce task stresses product grids, filters, browse density, cart/account conventions.

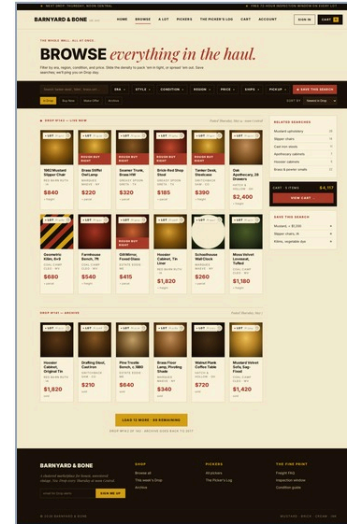
### Reference / oracle



trial reward 0.620 · coverage 0.882 · HTML 0.507 · CSSOM 0.589 · visual block 0.767

Agent gets the market feel but misses some density-specific states.

### Candidate



The marketplace task stresses browse grids, filter/control density, product cards, account/cart conventions, and state-specific density variants.

reward 0.620

coverage 0.882

HTML 0.507

CSSOM 0.589



# 7. Workflow task: booking state geometry

## Booking workflow exposes state/layout mistakes

Same contact step: text exists, but previous-step content remains and geometry changes.

### Reference / oracle

Meeples & Mug

BOARD GAME CAFE

Home

Menu

Library

Events

Book

Tavern Pass

Reviews

Book a table

1 Pick room

2 Date & party

3 Add-ons

4 Contact & deposit

Step 4 · Contact & deposit

Your name

Riley Tabletop

Phone

(555) 100-2000

Email

riley@example.com

Occasion (optional)

Friends night

Anything we should know?

Dietary needs, surprise guest, song requests...

☐ I agree to the deposit terms. The deposit applies to my final tab; refundable up to 48h before.

Back

Pay deposit & reserve

Your reservation

Room

not selected

When

2026-05-23 3h

Party

8 players

Estimated total

\$0

Deposit due

\$0

Deposit applies to your final tab. Free cancellation 48h ahead.

capture score 0.534 · HTML 0.613 · VB text 0.939 · VB size 0.061 · BBox 0.026 · CSSOM 0.047  
The task includes workflow states, not only polished landing pages.

### Candidate

Meeples & Mug

BOARD GAME CAFE

Home

Menu

Library

Events

Book

Tavern Pass

Reviews

Book a table

Step 3 · Add-ons (optional)

Tap to toggle. We'll add to the running total in the sidebar.

Snack platter

Hummus, fries, olives, pretzels — feeds 6. +\$24

Add

Dedicated host

A staff member explains rules, restricts dice, hands your group. +\$45

Add

Themed setup

We pre-arrange games on your theme: heist, fantasy, hidden ID. +\$15

Add

Back

Next contact

Step 4 · Contact & deposit

Your name

Riley Tabletop

Phone

(555) 100-2000

Email

riley@example.com

Occasion (optional)

Friends night

The tabletop task shows why multi-state tasks matter. The candidate includes the contact form, but keeps previous-step content and changes the page geometry.

capture score 0.534

VB text 0.939

VB size 0.061

BBox 0.026

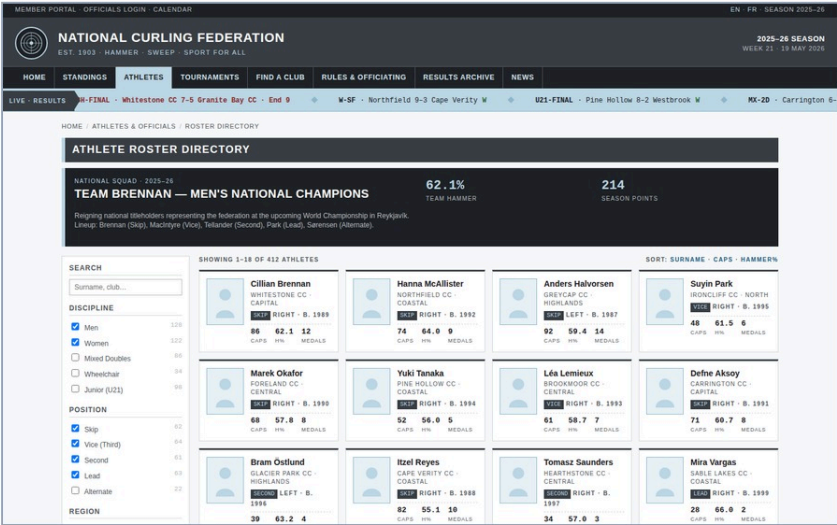


# 8. Sports task: local density is hard

## Sports federation: same domain, wrong density

Athlete directory reveals card density, table hierarchy, and nav/header geometry drift.

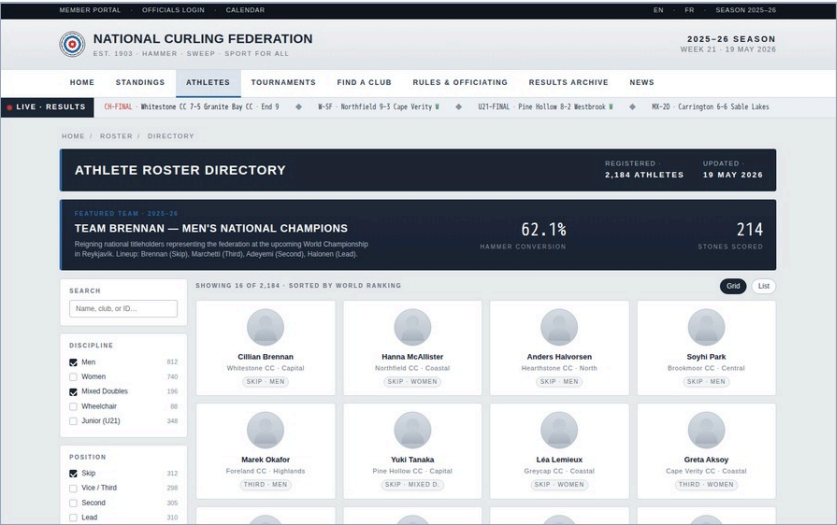
### Reference / oracle



trial reward 0.414 · athletes capture 0.218 · HTML 0.346 · BBox 0.119 · CSSOM 0.369

The agent understands the domain but redesigns the local system.

### Candidate



The agent recognizes the federation/roster domain, but redesigns card density, header geometry, footer/page proportions, and information hierarchy.

reward 0.414

athletes capture 0.218

HTML 0.346

BBox 0.119

# 9. Low-end example: plausible mood, weak fidelity

## Craft guild: broad mood is easier than exact grammar

Parchment mood survives, but typography hierarchy, copy density, and local structure drift.

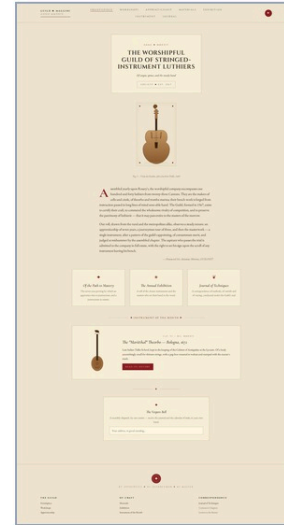
### Reference / oracle



trial reward 0.194 · DreamSim 0.849 · HTML 0.279 · CSSOM 0.271 · BBox 0.143

This is a useful low-end example: plausible aesthetic, weak faithful reproduction.

### Candidate



The lutherie task is useful because broad mood is preserved while exact typography, copy structure, and geometry are not. It shows why a task set needs stricter local metrics.

**reward** 0.194

**DreamSim** 0.849

**HTML** 0.279

**CSSOM** 0.271

# 10. Looks right at a glance, but structure drifts

## Indie record label: visually plausible, structurally weak

Music/editorial vibe is strong, while rendered content and page-specific structure drift.

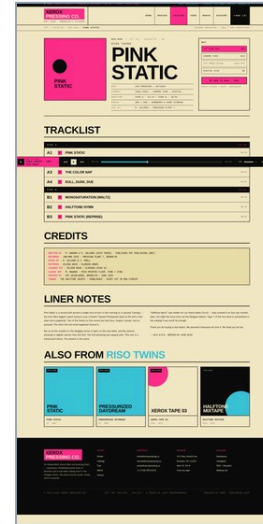
### Reference / oracle



trial reward 0.261 · VLM 0.652 · visual block 0.768 · HTML 0.314 · CSSOM 0.420

The benchmark separates genre-correct pages from faithful screenshots-to-code.

### Candidate



The indie-label task shows a common model behavior: produce a compelling genre page, but miss rendered content structure and local page-specific details.

**reward** 0.261

**VLM** 0.652

**visual block** 0.768

**HTML** 0.314

# 11. What this demonstrates

## Task quality

- Every task has at least 5 pages, and most have 10+ screenshots.
- State coverage makes tasks richer than static page matching.
- The hidden verifier boundary is clean: screenshot-only for the agent, oracle assets only for the verifier.
- The set covers institutional, editorial, commerce, nonprofit, media, sports, science, and operational UI.

## Agent behavior

- Strong at broad visual identity and page scaffolding.
- Weaker at exact density, CSS geometry, custom diagrams, and repeated component consistency.
- Weakest at alternate states and specific interaction captures.
- The reward spectrum broadly matches human intuition: better reproductions score higher, plausible rewrites score lower.

**Final read** This task set is a credible first dataset for the work trial: diverse generated websites, clean Harbor packaging, screenshot-only agent inputs, hidden replay evidence, and a reward surface that exposes meaningful reproduction failures.